

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import plotly.express as px
%matplotlib inline
```

```
df = pd.read_csv(
    "zomato.csv",
    engine="python",
    on_bad_lines="skip"
)
df
```

url	address	name	online_order	book_table	rate	votes	phone
https://www.zomato.com/bangalore/jalsa-banasha...	942, 21st Main Road, 2nd Stage, Banashankari, ...	Jalsa	Yes	Yes	4.1/5	775	080 42297555\r\n+91 9743772233
https://www.zomato.com/bangalore/spice-elephan...	2nd Floor, 80 Feet Road, Near Big Bazaar, 6th ...	Spice Elephant	Yes	No	4.1/5	787	080 41714161
s://www.zomato.com/SanchurroBangalore?cont...	1112, Next to KIMS Medical College, 17th Cross...	San Churro Cafe	Yes	No	3.8/5	918	+91 9663487993
https://www.zomato.com/bangalore/addhuri-udupi...	1st Floor, Annakuteera, 3rd Stage, Banashankar...	Addhuri Udupi Bhojana	No	No	3.7/5	88	+91 9620009302
https://www.zomato.com/bangalore/grand-village...	10, 3rd Floor, Lakshmi Associates, Gandhi Baza...	Grand Village	No	No	3.8/5	166	+91 8026612447\r\n+91 9901210005
...	...	...	...	...	...	...	...
os://www.zomato.com/bangalore/sea-spice-by-...	1890, Shetty Gardenia, 9th Main, Kaverinagar, ...	Sea Spice by 7 Star	Yes	Yes	4.0/5	349	080 49653393
https://www.zomato.com/bangalore/mini-punjabi-...	7/1, Behind BMS College Hostel, Hanumantha Nag...	Mini Punjabi Dhaba	Yes	No	4.2/5	287	+91 9845044074\r\n\r\n\r\n+91 7406012344
https://www.zomato.com/bangalore/onesta-banash...	2469, 3rd Floor, 24th Cross, Opposite BDA Comp...	Onesta	Yes	Yes	4.6/5	2556	080 48653961\r\n\r\n\r\n\r\n080 48655715
ps://www.zomato.com/bangalore/wamama-banash...	2466, Bhaskar Grand Tower, 24th Cross, Opposit...	Wamama	Yes	Yes	4.2/5	354	+91 8042111599\r\n\r\n\r\n\r\n+91 8971818009
https://www.zomato.com/bangalore/jalsa-banasha...	942, 21st Main Road, 2nd Stage, Banashankari, ...	Jalsa	Yes	Yes	4.1/5	775	080 42297555\r\n\r\n\r\n\r\n+91 9743772233

17 columns

Next steps:

[Generate code with df](#)[New interactive sheet](#)

df.columns

```
Index(['url', 'address', 'name', 'online_order', 'book_table', 'rate', 'votes',
      'phone', 'location', 'rest_type', 'dish_liked', 'cuisines',
      'approx_cost(for two people)', 'reviews_list', 'menu_item',
      'listed_in(type)', 'listed_in(city)'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 560 entries, 0 to 559
Data columns (total 17 columns):
#   Column                                Non-Null Count  Dtype
---  ---                                -
0   url                                    560 non-null    object
1   address                              560 non-null    object
2   name                                 560 non-null    object
3   online_order                         560 non-null    object
4   book_table                           560 non-null    object
5   rate                                 508 non-null    object
6   votes                                560 non-null    int64
7   phone                                554 non-null    object
8   location                             560 non-null    object
9   rest_type                           560 non-null    object
10  dish_liked                           253 non-null    object
11  cuisines                             558 non-null    object
12  approx_cost(for two people)          560 non-null    object
13  reviews_list                         560 non-null    object
14  menu_item                            560 non-null    object
15  listed_in(type)                      560 non-null    object
16  listed_in(city)                      560 non-null    object
dtypes: int64(1), object(16)
memory usage: 74.5+ KB
```

```
df.isnull().sum()
```

	0
url	0
address	0
name	0
online_order	0
book_table	0
rate	52
votes	0
phone	6
location	0
rest_type	0
dish_liked	307
cuisines	2
approx_cost(for two people)	0
reviews_list	0
menu_item	0
listed_in(type)	0
listed_in(city)	0

```
dtype: int64
```

```
df.drop(['url', 'address', 'phone', 'menu_item', 'dish_liked', 'reviews_list', 'listed_in(city)'], axis=1, inplace=True)
df.head(5)
```

	name	online_order	book_table	rate	votes	location	rest_type	cuisines	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1/5	775	Banashankari	Casual Dining	North Indian, Mughlai, Chinese	800	Buffet
1	Spice Elephant	Yes	No	4.1/5	787	Banashankari	Casual Dining	Chinese, North Indian, Thai	800	Buffet
2	San Churro Cafe	Yes	No	3.8/5	918	Banashankari	Cafe, Casual Dining	Cafe, Mexican, Italian	800	Buffet
3	Addhuri Idli	No	No	3.7/5	88	Banashankari	Quick Bites	South Indian,	300	Buffet

Next steps: [Generate code with df](#) [New interactive sheet](#)

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 560 entries, 0 to 559
Data columns (total 10 columns):
#   Column              Non-Null Count  Dtype
---  ---
0   name                 560 non-null    object
1   online_order         560 non-null    object
2   book_table          560 non-null    object
3   rate                 508 non-null    object
4   votes                560 non-null    int64
5   location             560 non-null    object
6   rest_type            560 non-null    object
7   cuisines             558 non-null    object
8   approx_cost(for two people) 560 non-null    object
9   listed_in(type)      560 non-null    object
dtypes: int64(1), object(9)
memory usage: 43.9+ KB
```

df.isnull()

	name	online_order	book_table	rate	votes	location	rest_type	cuisines	approx_cost(for two people)	listed_in(type)
0	False	False	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...	...	...
555	False	False	False	False	False	False	False	False	False	False
556	False	False	False	False	False	False	False	False	False	False
557	False	False	False	False	False	False	False	False	False	False
558	False	False	False	False	False	False	False	False	False	False
559	False	False	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...	...	...

df.isnull().sum()

	0
name	0
online_order	0
book_table	0
rate	52
votes	0

```
df.rename(columns={'name':'restaurants','book_table':'booking','rate':'rating','approx_cost(for two people)':'cost','listed_in('
df.head(5)
```

	restaurants	online_order	booking	rating	votes	location	rest_type	cuisines	cost	types	
0	Jalsa	Yes	Yes	4.1/5	775	Banashankari	Casual Dining	North Indian, Mughlai, Chinese	800	Buffet	
1	Spice Elephant	Yes	No	4.1/5	787	Banashankari	Casual Dining	Chinese, North Indian, Thai	800	Buffet	
2	San Churro Cafe	Yes	No	3.8/5	918	Banashankari	Cafe, Casual Dining	Cafe, Mexican, Italian	800	Buffet	

Next steps:

[Generate code with df](#)[New interactive sheet](#)

len(df)

560

df.dropna(inplace=True)

df.isnull().values.any()

np.False_

df.info()

```
<class 'pandas.core.frame.DataFrame'>
Index: 507 entries, 0 to 559
Data columns (total 10 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   restaurants      507 non-null    object
1   online_order     507 non-null    object
2   booking          507 non-null    object
3   rating           507 non-null    object
4   votes            507 non-null    int64
5   location         507 non-null    object
6   rest_type        507 non-null    object
7   cuisines         507 non-null    object
8   cost             507 non-null    object
9   types            507 non-null    object
dtypes: int64(1), object(9)
memory usage: 43.6+ KB
```

df[df.duplicated()].count().sum()

np.int64(0)

df.drop_duplicates(inplace=True)

df[df.duplicated()].count().sum()

np.int64(0)

df['rating'].unique()

```
array(['4.1/5', '3.8/5', '3.7/5', '3.6/5', '4.6/5', '4.0/5', '4.2/5',
       '3.9/5', '3.1/5', '3.0/5', '3.2/5', '3.3/5', '2.8/5', '4.4/5',
       '4.3/5', 'NEW', '2.9/5', '3.5/5', '2.6/5', '3.8 /5', '3.4/5',
       '4.5/5', '2.5/5', '2.7/5'], dtype=object)
```

```
df['rating'].replace("NEW|-", '0', regex=True).replace('/', '5', '', regex=True).unique()
```

```
array(['4.1', '3.8', '3.7', '3.6', '4.6', '4.0', '4.2', '3.9', '3.1',  
      '3.0', '3.2', '3.3', '2.8', '4.4', '4.3', '0', '2.9', '3.5', '2.6',  
      '3.8', '3.4', '4.5', '2.5', '2.7'], dtype=object)
```

```
df['rating'].isnull().sum()
```

```
np.int64(0)
```

```
df['location'].unique()
```

```
array(['Banashankari', 'Basavanagudi', 'Mysore Road', 'Jayanagar',  
      'Kumaraswamy Layout', 'Rajarajeshwari Nagar', 'Vijay Nagar',  
      'Uttarahalli', 'JP Nagar', 'South Bangalore', 'City Market'],  
      dtype=object)
```

```
df['location'].isnull().sum()
```

```
np.int64(0)
```

```
df['cost'].unique()
```

```
array(['800', '300', '600', '700', '550', '500', '450', '650', '400',  
      '900', '200', '750', '150', '850', '100', '1,200', '350', '250',  
      '950', '1,000', '1,500', '1,300', '199'], dtype=object)
```

```
df['cost'].isnull().sum()
```

```
np.int64(0)
```

```
df['cost'] = df['cost'].replace(",", '', regex=True).astype(int)
```

```
df.cost.unique()
```

```
array([ 800,  300,  600,  700,  550,  500,  450,  650,  400,  900,  200,  
        750,  150,  850,  100, 1200,  350,  250,  950, 1000, 1500, 1300,  
        199])
```

```
df['booking'].unique()
```

```
array(['Yes', 'No'], dtype=object)
```

```
df['booking'].isnull().sum()
```

```
np.int64(0)
```

```
df['online_order'].unique()
```

```
array(['Yes', 'No'], dtype=object)
```

```
df['online_order'].isnull().sum()
```

```
np.int64(0)
```

```
df['rest_type'].unique()
```

```
array(['Casual Dining', 'Cafe, Casual Dining', 'Quick Bites',  
      'Casual Dining, Cafe', 'Cafe', 'Quick Bites, Cafe',  
      'Cafe, Quick Bites', 'Delivery', 'Mess', 'Dessert Parlor',  
      'Bakery, Dessert Parlor', 'Pub', 'Bakery', 'Takeaway, Delivery',  
      'Fine Dining', 'Beverage Shop', 'Sweet Shop', 'Bar',  
      'Dessert Parlor, Sweet Shop', 'Bakery, Quick Bites',  
      'Sweet Shop, Quick Bites', 'Kiosk', 'Food Truck',  
      'Quick Bites, Dessert Parlor', 'Beverage Shop, Quick Bites',  
      'Beverage Shop, Dessert Parlor', 'Takeaway', 'Pub, Casual Dining',  
      'Casual Dining, Bar', 'Dessert Parlor, Beverage Shop',  
      'Quick Bites, Bakery', 'Microbrewery, Casual Dining'], dtype=object)
```

```
b = df['rest_type'].value_counts()  
b
```

	count
rest_type	
Quick Bites	208
Casual Dining	104
Cafe	49
Dessert Parlor	36
Bakery	16
Takeaway, Delivery	12
Delivery	12
Sweet Shop	12
Beverage Shop	9
Casual Dining, Cafe	5
Bar	4
Cafe, Casual Dining	4
Beverage Shop, Quick Bites	4
Sweet Shop, Quick Bites	4
Casual Dining, Bar	3
Beverage Shop, Dessert Parlor	3
Bakery, Quick Bites	3
Cafe, Quick Bites	2
Quick Bites, Cafe	2
Bakery, Dessert Parlor	2
Dessert Parlor, Sweet Shop	2
Fine Dining	1
Mess	1
Pub	1
Quick Bites, Dessert Parlor	1
Food Truck	1
Kiosk	1
Takeaway	1
Pub, Casual Dining	1
Dessert Parlor, Beverage Shop	1
Quick Bites, Bakery	1
Microbrewery, Casual Dining	1

dtype: int64

```
df['types'].unique()
```

```
array(['Buffet', 'Cafes', 'Delivery', 'Desserts', 'Dine-out'],
      dtype=object)
```

```
df['types'].value_counts()
```

```
count
types
Delivery    418
Desserts     51
Cafes        24
Buffet        7
Dine-out     7
```

```
dtype: int64
```

```
df['cuisines'].isnull().sum()
```

```
np.int64(0)
```

```
df['cuisines'].value_counts()
```

	count
cuisines	
South Indian	32
North Indian	22
North Indian, Chinese	20
Bakery, Desserts	13
Fast Food	13
...	...
Tea, Beverages	1
Beverages, Juices, Desserts, Italian	1
Mithai, Street Food, North Indian, Chinese	1
Bakery, Street Food	1
Continental, Asian, North Indian, Tea	1

```
244 rows × 1 columns
```

```
dtype: int64
```

```
df['restaurants'].unique()
```

```
array(['Jalsa', 'Spice Elephant', 'San Churro Cafe',
      'Addhuri Udupi Bhojana', 'Grand Village', 'Timepass Dinner',
      'Rosewood International Hotel - Bar & Restaurant', 'Onesta',
      'Penthouse Cafe', 'Smaczne',
      'CafÃ\x83Ã\x82Ã\x83Ã\x82Ã\x82Ã\x83Ã\x83Ã\x82Ã\x82Ã\x83Ã\x82Ã\x82Ã Down The Alley',
      'Cafe Shuffle', 'The Coffee Shack', 'Caf-Eleven', 'Cafe Vivacity',
      'Catch-up-ino', 'Kirthi's Biryani', 'T3H Cafe',
      '360 Atoms Restaurant And Cafe', 'The Vintage Cafe',
      'Woodee Pizza', 'Cafe Coffee Day', 'My Tea House', 'Hide Out Cafe',
      'CAFE NOVA', 'Coffee Tindi', 'Sea Green Cafe', 'Cuppa',
      'Srinathji's Cafe', 'Redberrys', 'Foodiction', 'Sweet Truth',
      'Ovenstory Pizza', 'Faasos', 'Behrouz Biryani', 'Fast And Fresh',
      'Szechuan Dragon', 'Empire Restaurant',
      'Maruthi Davangere Benne Dosa', 'Chaatimes', 'Havyaka Mess',
      'McDonald's', 'Domino's Pizza', 'Hotboxit', 'Kitchen Garden',
      'Recipe', 'Beijing Bites', 'Tasty Bytes', 'Petoo',
      'Shree Cool Point', 'Corner House Ice Cream', 'Biryani's And More',
      'Roving Feast', 'FreshMenu', 'Banashankari Donne Biryani',
      'Wamama', 'Five Star Chicken', 'XO Belgian Waffle',
      'Peppy Peppers', 'Goa 0 Km', 'Chinese Kitchen', '1947',
      'Cake of the Day', 'Kabab Magic', 'Namma Brahmin's Idli',
      'Gustoes Beer House', 'Sugar Rush', 'Burger King', 'The Good Bowl',
      'The Biryani Cafe', 'Spicy Tandoor', 'LSD Cafe', 'Rolls On Wheels',
      'Om Sri Vinayaka Chats',
      'Sri Guru Kottureshwara Davangere Benne Dosa',
      'Devanna Dum Biryani Centre', 'Kolbeh', 'Upahar Sagar',
      'Kadalu Sea Food Restaurant', 'Frozen Bottle', 'Parimala Sweets',
      'Vaishali Deluxe', 'The Big O Bakes', 'Meghana Foods',
      'Krishna Sagar', 'Dessert Rose', 'Chickpet Donne Biryani House',
```



```
"Thanco's Natural Ice Creams", 'Nandhini Deluxe',
"Vi Ra's Bar and Restaurant", 'Kaggis', 'Ayda Persian Kitchen',
'Chatar Patar', 'Polar Bear', 'Kidambi's Kitchen', 'Mane Thindi',
'Kotian Karavali Restaurant', 'Floured-Baked With Love',
'Cakes & Slices', 'Spice 9', 'Naveen Kabab & Biryani Mane',
'Katriguppe Donne Biryani', 'Hari Super Sandwich',
'Atithi Point Ande Ka Funda', 'Just Bake', 'K27 - The Pub',
'Bengaluru Coffee House', 'New Mangalore Lunch Home',
'Coffee Bytes', 'Parjanya Chat Zone',
"Kwality Wall's Swirl's Happiness Station", 'Ruchi Maayaka',
'Roll Magic Fast Food', 'Foodlieious Multi Cuisine',
'Thanishka Nati And Karavali Style', 'Melting Melodies',
'New Indraprasta', 'Anna Kuteera', 'Darbar', 'Vijayalakshmi',
'Sri Udupi Food Hub', 'Udupi Upahar', 'House Of Kebabs',
'Roll N Rock', 'BOX8- Desi Meals', 'KFC', 'Bitez', 'Roll Over',
'Imperial Restaurant', 'Lassi Shop', 'The Fortuna', 'Wahab',
'Al Diwan Biryani's And More', 'New Gowda's Fried Chicken',
'Canon', 'Diabetics Deziere Sugarless Sweets and Bakes',
'The Blue Wagon - Kitchen', 'HOT COFFEE', 'Patio 805',
'Lassi Corner', 'Sagar Deluxe', 'Kanti Sweets', 'Vegetalia',
'Aramane Donne Biryani', 'Ande Ka Funda', 'Cake Ghar',
'Energy Addaa', 'Bhattara Bhojana', 'Tandoori Knight',
'Dev Sagar - Food Street', 'Mittraa Da Pizza', 'Paradise Premium',
'Grazers', 'Shakes Theory', '@Italy', 'Chilli Flakes',
'Calcutta Cafe', 'Old Mumbai Ice Cream', 'Donne Biryani House',
'By 2 Coffee', 'Kedia's Fun Food', 'Davangere Butter Dosa Hotel',
'Just Shawarma', 'Mini Punjabi Dhaba', 'Mulabagilu Dosa',
'Gokul Veg', 'Olive - Era', 'Udupi Ruchi',
'Medhara Hindu Militharu Hotel', 'Coco Cafe', 'Pizza Hut'
```

```
df.groupby('restaurants').count().head()
```

	online_order	booking	rating	votes	location	rest_type	cuisines	cost	types
restaurants									
1947	1	1	1	1	1	1	1	1	1
1980s Games Cafe	1	1	1	1	1	1	1	1	1
360 Atoms Restaurant And Cafe	2	2	2	2	2	2	2	2	2
@Italy	1	1	1	1	1	1	1	1	1
A2B - Adyar Ananda Bhavan	1	1	1	1	1	1	1	1	1

```
df['restaurants']=df['restaurants'].str.replace('[Ãx][^A-Za-z]+' ,'',regex=True)
```

```
df.groupby('restaurants').count().head()
```

	online_order	booking	rating	votes	location	rest_type	cuisines	cost	types
restaurants									
1947	1	1	1	1	1	1	1	1	1
1980s Games Cafe	1	1	1	1	1	1	1	1	1
360 Atoms Restaurant And Cafe	2	2	2	2	2	2	2	2	2
@Italy	1	1	1	1	1	1	1	1	1
A2B - Adyar Ananda Bhavan	1	1	1	1	1	1	1	1	1

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 507 entries, 0 to 559
Data columns (total 10 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   restaurants      507 non-null    object
1   online_order     507 non-null    object
2   booking          507 non-null    object
3   rating           507 non-null    object
4   votes            507 non-null    int64
5   location         507 non-null    object
6   rest_type       507 non-null    object
7   cuisines         507 non-null    object
8   cost             507 non-null    int64
9   types            507 non-null    object
dtypes: int64(2), object(8)
```

memory usage: 43.6+ KB

```
df.isnull().sum()
```

	0
restaurants	0
online_order	0
booking	0
rating	0
votes	0
location	0
rest_type	0
cuisines	0
cost	0
types	0

dtype: int64

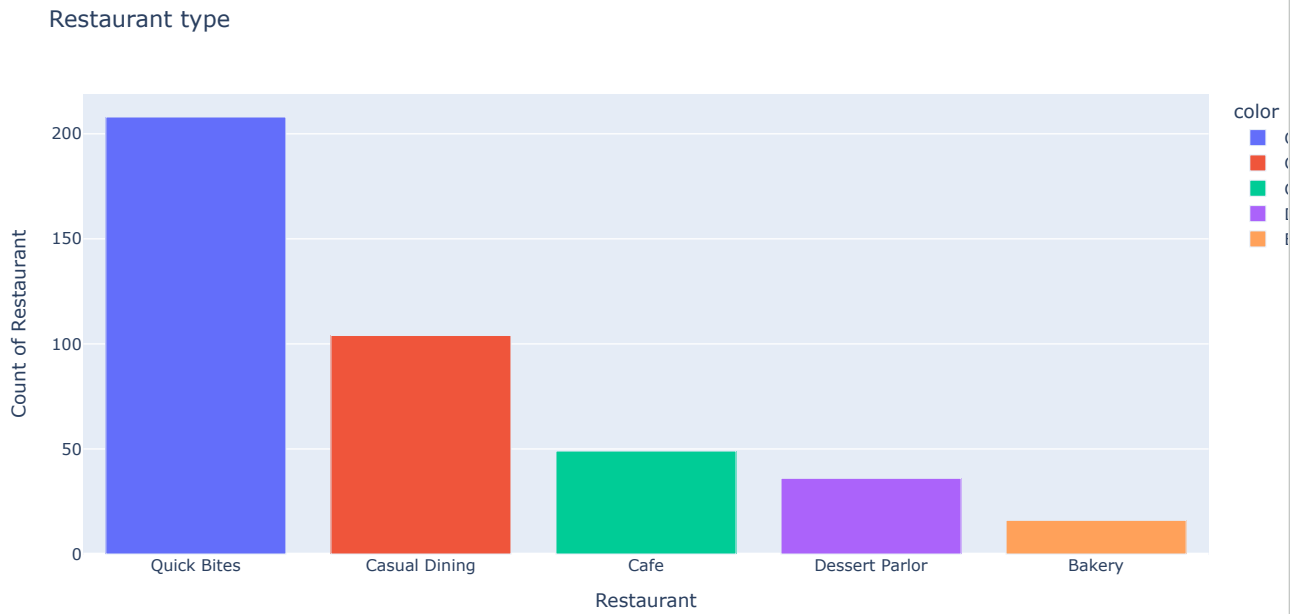
```
df.to_csv('./clean_zomato.csv')
```

```
type1=df['rest_type'].value_counts().head(5)  
type1.to_frame()
```

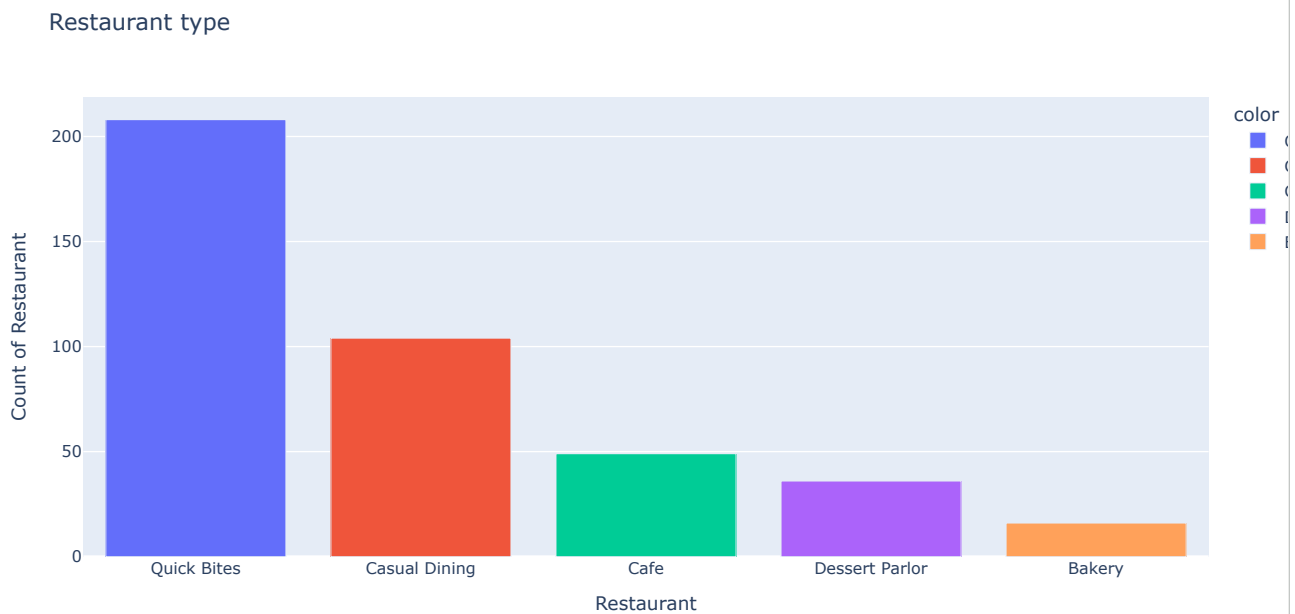
	count
rest_type	
Quick Bites	208
Casual Dining	104
Cafe	49
Dessert Parlor	36
Bakery	16

• Restaurant Type

```
fig = px.bar(x= type1.index,  
             y= type1,  
             color= type1.index,  
             title='Restaurant type',labels={'x':'Restaurant','y':'Count of Restaurant'})  
fig.show()
```



```
fig=px.bar(x=type1.index,  
           y=type1,  
           color=type1.index # Each bar is colored differently  
           ,title='Restaurant type',labels={'x':'Restaurant','y':'Count of Restaurant'}) # changes x and y label  
fig.show()
```



• Restaurants delivering Online or not

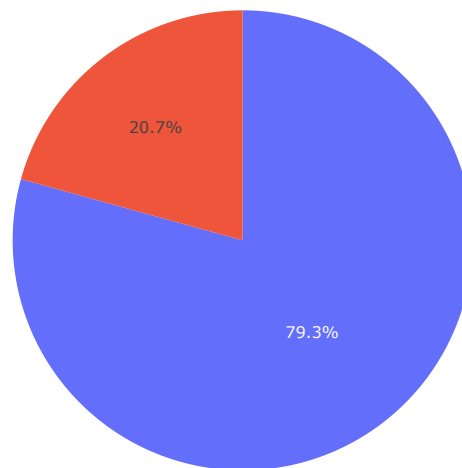
```
online_or_not = df['online_order'].value_counts()  
online_or_not
```

	count
online_order	
Yes	402
No	105

dtype: int64

```
fig2 = px.pie(online_or_not,  
              title = ' Delivering Online or Not',  
              values = online_or_not,  
              names = online_or_not.index  
              )  
fig2.show()
```

Delivering Online or Not



Double-click (or enter) to edit

•Restaurants allowing table booking or not

```
df['booking'].value_counts()
```

	count
booking	
No	471
Yes	36

dtype: int64

```
fig = px.bar(df,x='booking', color = 'booking',  
            ,title="table booking or not"  
            )  
fig.update_layout(  
    width=700, # 🖱️ change this  
    height=450 # 🖱️ and this  
)
```

```
fig.show()
```

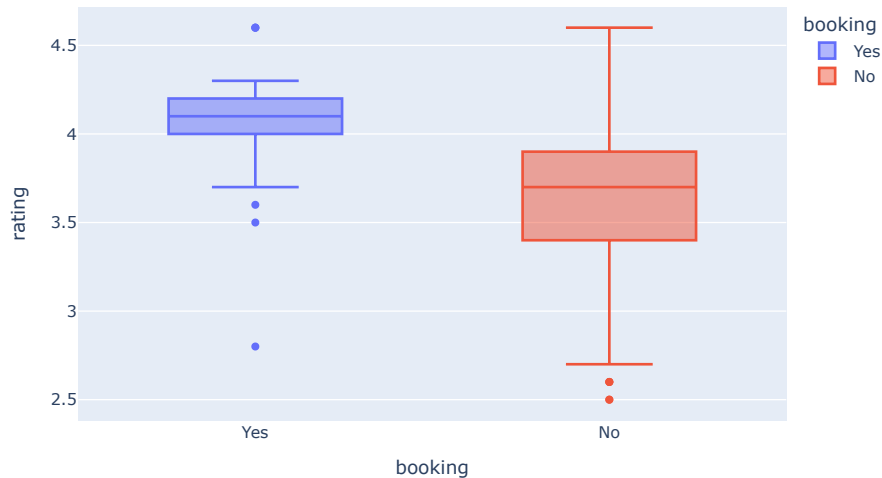
table booking or not



• Table booking Rate vs Rate

```
fig= px.box(df,x='booking',
            y='rating',
            color='booking'
            )
fig.update_layout(
    width=700, # 🍌 change this
    height=450 # 🍌 and this
)

fig.show()
```



Online Order vs Rate

```
fig = px.violin(df,x='online_order', y="rating",
                color='online_order',

                # draw box plot inside the violin
                points='outliers', # can be 'outliers', or False
                )
```

```
fig.update_layout(
    width=700, # 🖱️ change this
    height=450 # 🖱️ and this
)
```

```
fig.show()
```



• Best Location

```
loc_vote = df.groupby('location')['votes'].sum()
loc_vote = loc_vote.nlargest(5).sort_values()
loc_vote
```

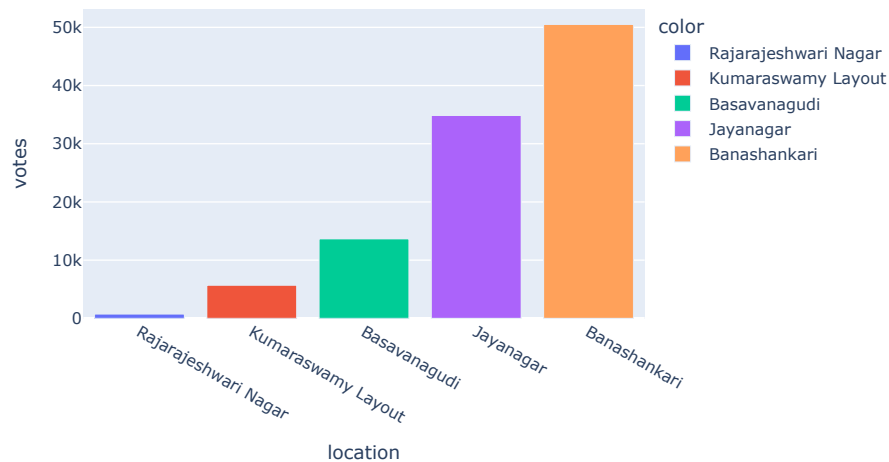
	votes
location	
Rajarajeshwari Nagar	732
Kumaraswamy Layout	5676
Basavanagudi	13655
Jayanagar	34854
Banashankari	50496

dtype: int64

```
fig = px.bar(x=loc_vote.index,
             y=loc_vote,
             color=loc_vote.index,
             title='Best Location', labels={'x': 'location', 'y': 'votes'})
fig.update_layout(
    width=700, # 🖱️ change this
    height=450 # 🖱️ and this
)

fig.show()
```

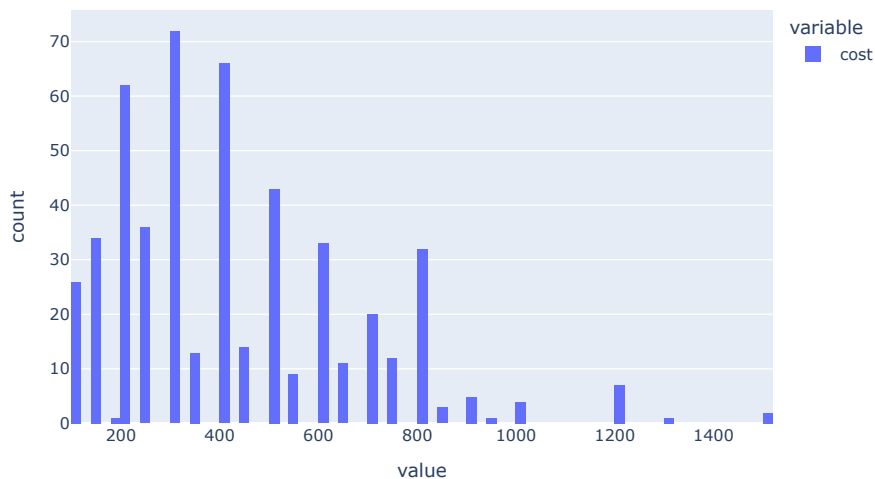
Best Location



• Cost of Restaurant

```
fig = px.histogram(df['cost'],nbins=100)
fig.update_layout(
    width=700, # 🐞 change this
    height=450 # 🐞 and this
)

fig.show()
```



```
df['rating'] = (
    df['rating']
    .astype(str)
    .str.split('/')
    .str[0]
)

df['rating'] = pd.to_numeric(df['rating'], errors='coerce')
```

```
vote = (
    df.groupby('restaurants', as_index=False)
    .agg({
```

```

        'votes': 'mean',
        'rating': 'mean'
    })
    .sort_values(by=['rating', 'votes'], ascending=False)
    .head(10)
)

```

Best Location by Votes

(Relation between Location and Votes)

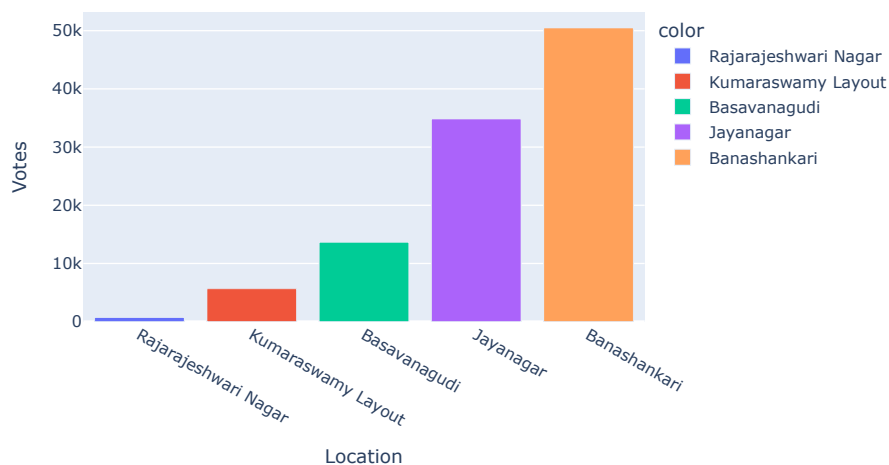
```

fig = px.bar(
    x=loc_vote.index,
    y=loc_vote.values,
    color=loc_vote.index,
    title='Best location by votes',
    labels={'x': 'Location', 'y': 'Votes'})
)
fig.update_layout(
    width=700, # 🖱️ change this
    height=450 # 🖱️ and this
)

fig.show()

```

Best location by votes



Best Location by Rating

(Relation between Location and Rating)

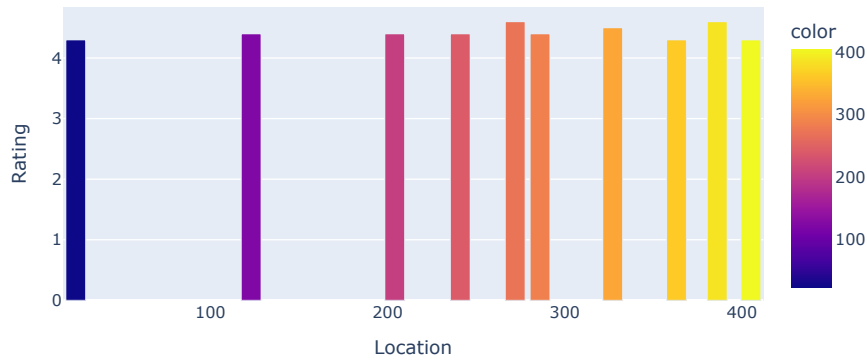
```

fig=px.bar(y=vote['rating'],
            x=vote.index,
            color=vote.index # Each bar is colored differently
            ,title='Best location by rating',labels={'x':'Location','y':'Rating'}) # changes x and y label
)
fig.update_layout(
    width=700, # 🖱️ change this
    height=400 # 🖱️ and this
)

fig.show()

```

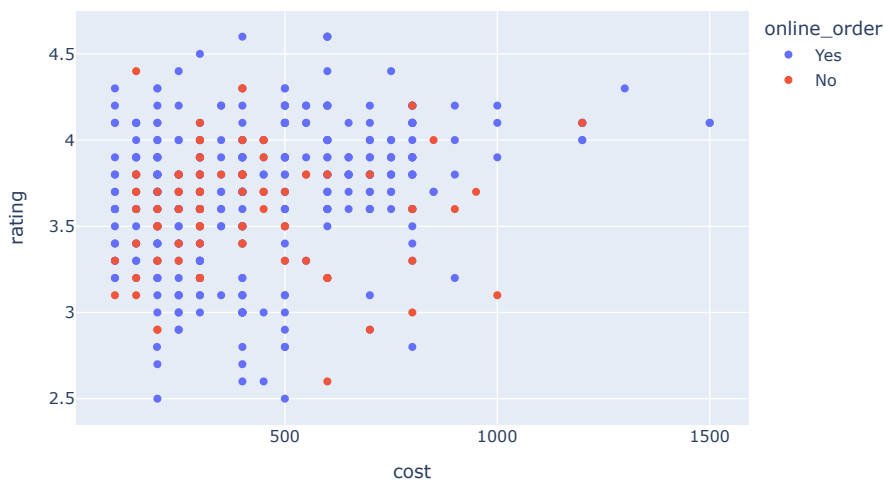

Best location by rating



Relation between cost and rating

```
fig=px.scatter(df,x='cost',y='rating',color='online_order')
fig.update_layout(
    width=700, # 🖱️ change this
    height=450 # 🖱️ and this
)

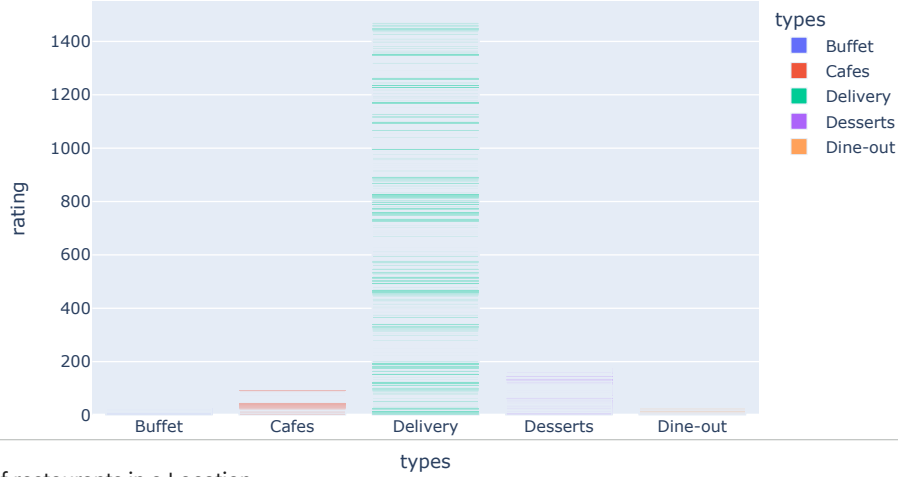
fig.show()
```



Relation between Location and Type

```
fig = px.bar(df,x="types",y ="rating",color='types')
fig.update_layout(
    width=700, # 🖱️ change this
    height=450 # 🖱️ and this
)

fig.show()
```



• No. of restaurants in a Location

df.head(5)

	restaurants	online_order	booking	rating	votes	location	rest_type	cuisines	cost	types	
0	Jalsa	Yes	Yes	4.1	775	Banashankari	Casual Dining	North Indian, Mughlai, Chinese	800	Buffet	
1	Spice Elephant	Yes	No	4.1	787	Banashankari	Casual Dining	Chinese, North Indian, Thai	800	Buffet	
2	San Churro Cafe	Yes	No	3.8	918	Banashankari	Cafe, Casual Dining	Cafe, Mexican, Italian	800	Buffet	